BAYESIAN MODELING OF COMPLEXITY

Bayesian Calibration

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Hertie School

Agenda

1 Recap & Today

2 Bayesian Calibration Methods

- Approximate Bayesian Computation
- Computational Limitation

3 Practical Calibration Workflow

4 Wrap-up & Next

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2 Bayesian Calibration Methods

3 Practical Calibration Workflow

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Where We Are in the Course

What we've covered:

- ✓ **Sessions 1-3:** ABM design, implementation, analysis
- ✓ **Sessions 4-6:** Bayesian inference fundamentals

Today: Bridge ABMs $\leftrightarrow$ Bayesian inference

The central question:

> Given observed data from a real system, what parameter values in our ABM best explain that data?

The ABM Calibration Challenge

Scenario: You've built a Schelling segregation model

- **Unknown:** Tolerance threshold θ (the parameter)
- **Known:** Real city segregation index $y^* = 0.78$ (the data)

Goal: Infer posterior $p(\theta \mid y^*)$

The problem: Standard Bayesian inference requires likelihood $p(y \mid \theta)$

For ABMs, we **cannot write down** $p(y \mid \theta)$ analytically!

- No closed-form formula
- Too many interacting stochastic processes
- Emergent outcomes from complex interactions

Key Challenge

ABMs are **simulators**, not probability distributions. We can generate data $y \sim \text{ABM}(\theta)$ but CANNOT evaluate $p(y \mid \theta)$.

Why Traditional Methods Fail

Standard Bayesian inference:

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

- MCMC: Evaluates $p(y \mid \theta)$ at each step
- Variational inference: Requires gradients of $\log p(y \mid \theta)$
- Both need explicit likelihood!

Maximum likelihood estimation:

$$\hat{\theta} = \arg\max_{\theta} p(y \mid \theta)$$

No uncertainty quantification!

For ABMs:

- We can simulate: $y_{\text{sim}} \sim \text{ABM}(\theta)$
- We cannot evaluate: $p(y \mid \theta) = ???$
- Stochasticity $\Rightarrow$ same θ gives different y
- Emergence $\Rightarrow$ no analytical relationship

What we need

Likelihood-free methods that use simulations instead of $p(y \mid \theta)$

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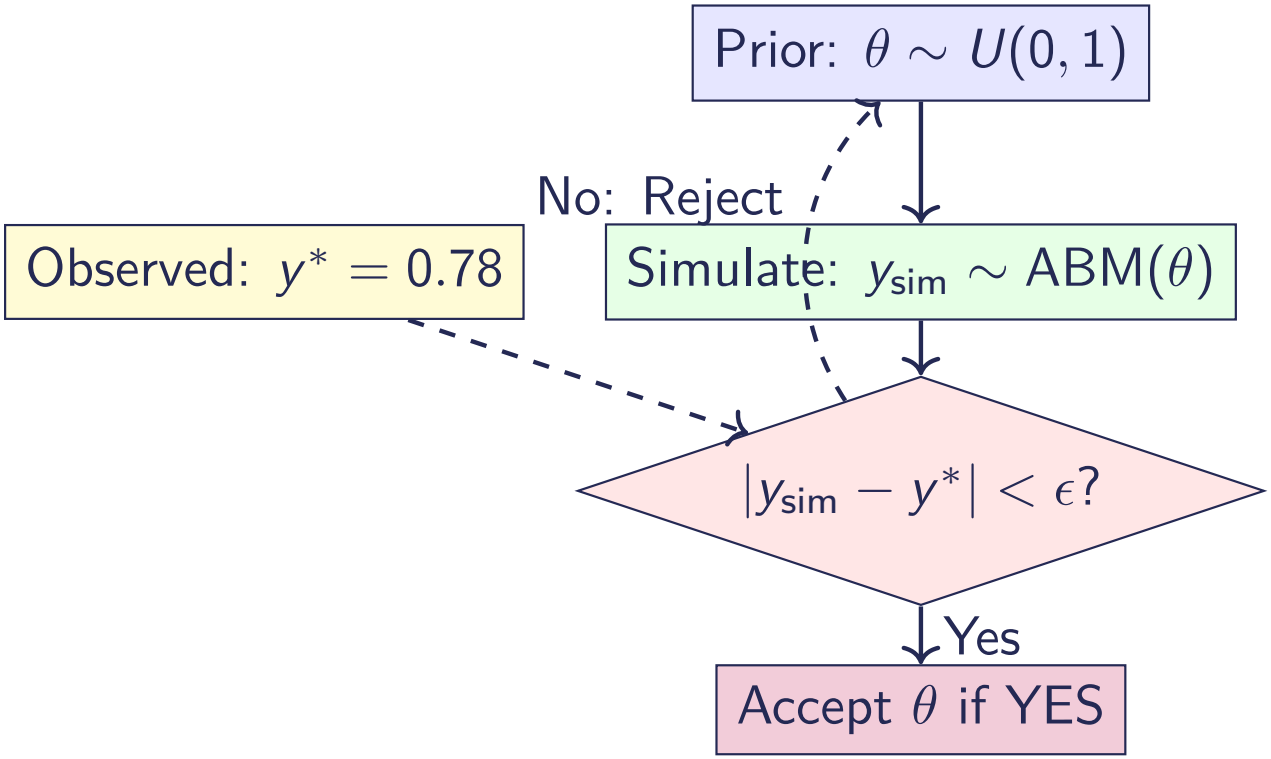
Approximate Bayesian Computation (ABC): The Core Idea

ABC: Bayesian inference without likelihoods

Logic: If simulation closely matches data, parameter value is plausible



Tolerance ϵ: How close is "close enough"?

ABC Rejection Algorithm

Algorithm 1 ABC Rejection Sampling

Input: Observed data y^*, prior $p(\theta)$, tolerance ϵ, ABM simulator
Initialize: $S = \emptyset$ (empty set of accepted parameters)
while $|S| < N$ (target number of samples) **do**
    Sample $\theta \sim p(\theta)$ from prior
    Simulate $y_{\text{sim}} \sim \text{ABM}(\theta)$
    Compute distance $d = \|y_{\text{sim}} - y^*\|$
    if $d < \epsilon$ **then**
        Add θ to S
    end if
end while
Output: $S \approx$ sample from $p(\theta \mid y^*)$

Key properties:

- As $\epsilon \to 0$, ABC posterior $\to$ true posterior
- smaller tolerance $\to$ more rejections
- Computationally expensive for tight tolerances

Distance Metrics and Summary Statistics

Problem: ABM outputs are high-dimensional, e.g., 50×50 grid

Solution: Use **summary statistics** $S(y)$ instead of raw output y

Summary Statistics

Low-dimensional representations capturing key features:

- Schelling: Seg. index, clustering coefficient, happiness rate
- Disease model: Peak time, total infections, duration
- Traffic: Average speed, variance, congestion patterns

Distance metrics:

- (Weighted) Euclidean: $d = \sqrt{\sum_i w_i (S_i(y_{\text{sim}}) - S_i(y^*))^2}$

Critical Choice

Summary stats must be **sufficient** (?): capture all info about θ in data

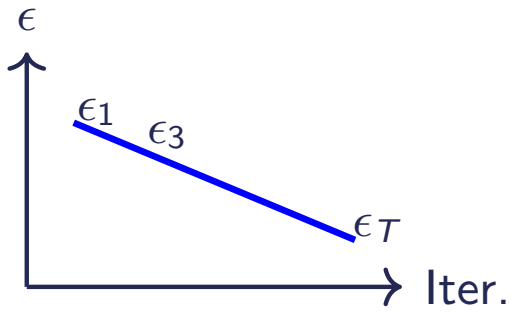
ABC Variants: Sequential Monte Carlo

Problem of rejection ABC: Simulation waste when ϵ is small

Solution: ABC-SMC (Sequential Monte Carlo ABC)

Idea: Start with loose tolerance, gradually tighten

1. Start: ϵ_1 large (easy to accept)
2. Get initial particle set from $p(\theta \mid d < \epsilon_1)$
3. Decrease tolerance: $\epsilon_2 < \epsilon_1$
4. Use previous particles as proposal distribution
5. Perturb & resample with importance weights
6. Repeat until ϵ_T is sufficiently small



Adaptive tolerance schedule

Advantages: Much more efficient, better exploration of posterior

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The Computational Bottleneck

Problem: ABMs are computationally expensive

Example: Schelling Model

- 1 simulation: 5 seconds
- ABC needs ∼10,000 simulations for good posterior
- Total: $5 \times 10{,}000 = 50{,}000$ seconds $\approx$ 14 hours!

For realistic ABMs:

- 1 simulation: minutes to hours
- Multiple parameters to calibrate
- Multiple patterns to match
- $\Rightarrow$ Infeasible for direct ABC!

Solution: Emulation

Replace expensive ABM with fast **surrogate model** (emulator)

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Calibration Scenario

Goal: Infer tolerance threshold from observed segregation

Setup:

- **Unknown parameter:** Tolerance $\theta \in [0, 1]$
- **Observed data:** Segregation index $y^* = 0.78$ (e.g., Chicago)
- **Prior:** $\theta \sim \text{Uniform}(0, 1)$ (uninformative)
- **ABM:** Schelling model (50×50 grid, 2000 agents, 80% occupancy)

Question: What tolerance value(s) best explain observed segregation?

Approach:

1. Run Schelling at 20 tolerance values (design points)
2. Use ABC with emulator to get posterior $p(\theta \mid y^* = 0.78)$
3. Validate on other patterns (convergence time, clustering)

Validation is Critical!

Calibration $\neq$ Validation

Validation Strategy

1. **Within-sample:** Do calibrated parameters reproduce training patterns?
2. **Held-out patterns:** Do they reproduce patterns *not* used in calibration?
3. **Posterior predictive:** Sample many θ from posterior, check distribution of outputs
4. **Out-of-sample:** New data (different time, location, etc.)

Example: Schelling

- **Calibrate to:** Final segregation index
- **Validate on:** Spatial clustering, happiness
- If validation fails $\Rightarrow$ restructure your model

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Key Takeaways

1. **ABM calibration requires likelihood-free methods**
   - Cannot write $p(y \mid \theta)$ analytically
   - Use simulations instead of likelihoods
2. **ABC: Core method for ABM inference**
   - Accept parameters if simulations close to data
   - Variants: Rejection, SMC, MCMC etc.
3. **Emulators dramatically reduce computational cost**
   - GPs are the most common
   - Neural networks for high dimensions
4. **Validation is essential**
   - Calibration $\neq$ validation
   - Check multiple patterns (POM!)
   - Posterior predictive checks

For Your ABM Projects

What you should do:

1. **Identify calibration targets:** What patterns will you match?
2. **Decide on approach:**
   - Simple ABC
   - ABC-SMC
   - Emulator (advanced)
3. **Start with exploration:** Parameter sweeps (Session 3 material)
4. **Implement basic ABC:** Rejection sampler first
5. **Validate carefully:** Multiple patterns, posterior predictive checks
6. **Report uncertainty:** Posterior distributions, not just point estimates

Questions?

Next session: Bayesian Linear Models